

Leveraging AI to Support Problem Representation Instruction

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Today's Agenda



Proof of Concept

Sharing our workflow for developing an AI-supported PR learning supplement using prompting and Shiny.

-  **1** What is Problem Representation (PR)?
-  **2** Why does PR matter?
-  **3** What makes an effective PR?
-  **4** What we know about PR instruction
-  **5** Building a PR learning supplement with AI + Shiny
-  **6** Demo

What is Problem Representation (PR)?

(aka “one-liner”, “the summary statement”, “the assessment”)

Dynamic mental representation of the relevant aspects of a case that can be communicated in a concise summary incorporating key findings and semantic qualifiers

**Information
Gathering**

**Hypothesis
Generation**

**Problem
Representation**

**Differential
Diagnosis**

**Leading/Working
Diagnosis**

**Diagnostic
Justification**

**Management and
Treatment**

Why is PR important?

Acute abdominal pain

Poss ddx: gastrointestinal obstruction, inflammatory process, ischemic process, infectious process, urologic process



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Acute right lower quadrant abdominal pain

Poss ddx: **acute appendicitis**, diverticulitis, ureterolithiasis

Why is PR important?

Acute abdominal pain

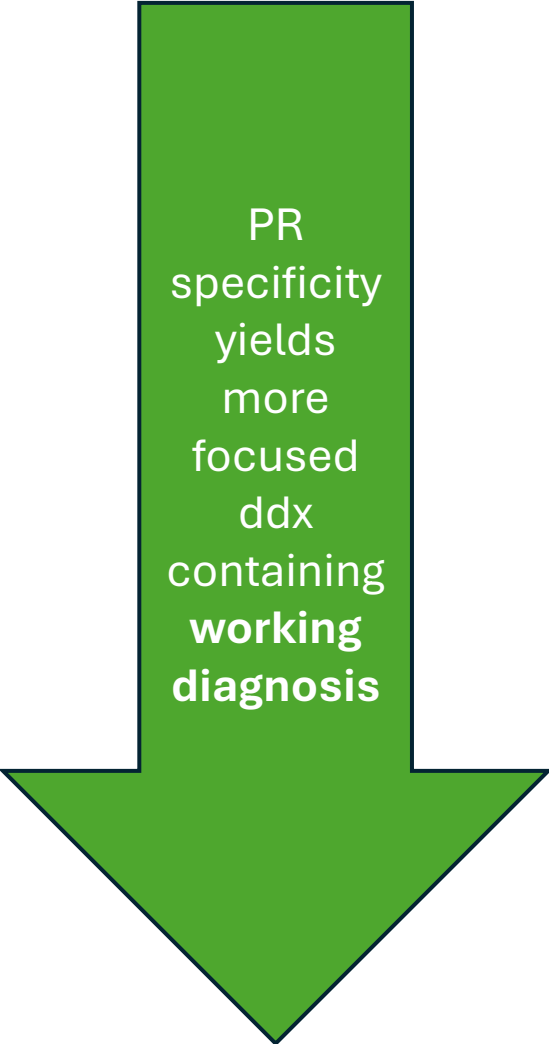
Poss ddx: gastrointestinal obstruction, inflammatory process, ischemic process, infectious process, urologic process

Acute right lower quadrant abdominal pain

Poss ddx: **acute appendicitis**, diverticulitis, ureterolithiasis

Acute, progressive right lower quadrant pain with fever and nausea

Poss ddx: **acute appendicitis**, appendiceal perforation, intra-abdominal abscess



PR
specificity
yields
more
focused
ddx
containing
**working
diagnosis**

What makes an effective PR?

*“A 23-year-old man with a first-degree family history of appendicitis (father) presents with acute onset, progressively worsening right lower quadrant abdominal pain over 14 hours, associated with fever, nausea, and anorexia.”

- Common Elements
 - Risk Factors
 - Time Course of Illness
 - Description of Clinical Syndrome

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Evidence Informing the Development of Our PR Learning Supplement

Multiple sources of evidence converged on the need for an interactive, feedback-rich tool

1. LITERATURE: CURRICULAR CONTEXT & INSTRUCTIONAL STRATEGIES



Curricular Context

Limited curricular time to explicitly teach Clinical Reasoning



Instructional Approaches

Effective approaches include:



Immediate feedback



Self-explanation & reflection



Retrieval practice



Case contrasting

2. STAKEHOLDER INSIGHTS



SME Focus Groups / Working Groups



Summer 2024
(3 meetings)



Focused on Clinical Reasoning Assessment & Feedback



PR is regarded as an integral part of Clinical Reasoning



Student Focus Groups



Spring 2025
(3 meetings)



Generally unfamiliar with PR (language or concept)



When definition was explained, they saw value



10 students



DESIGN RATIONALE

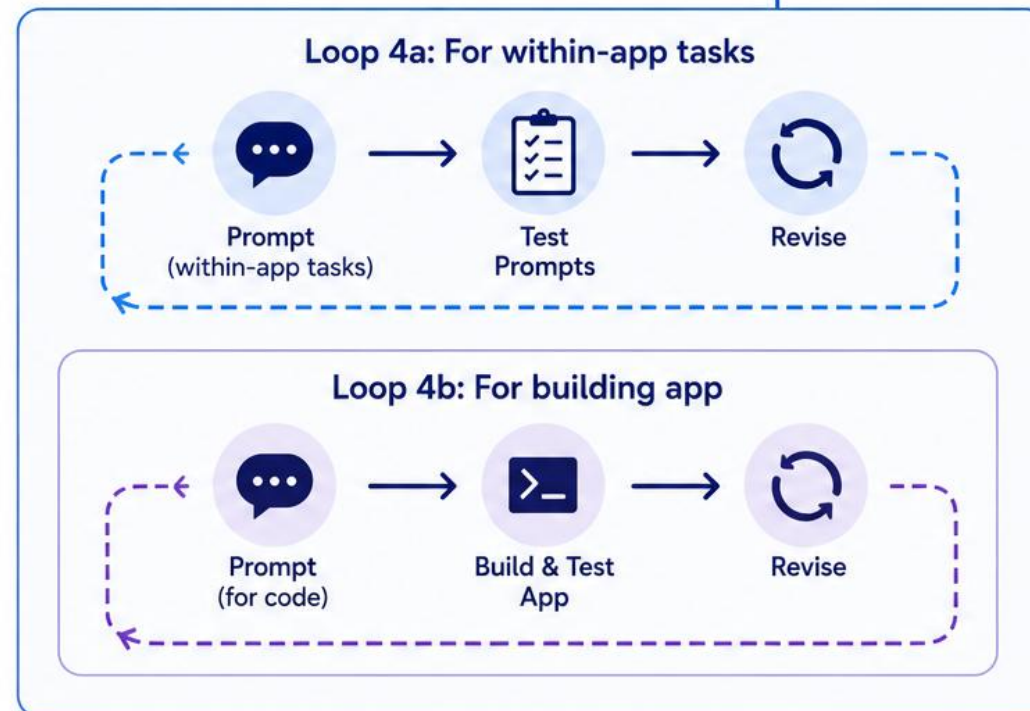
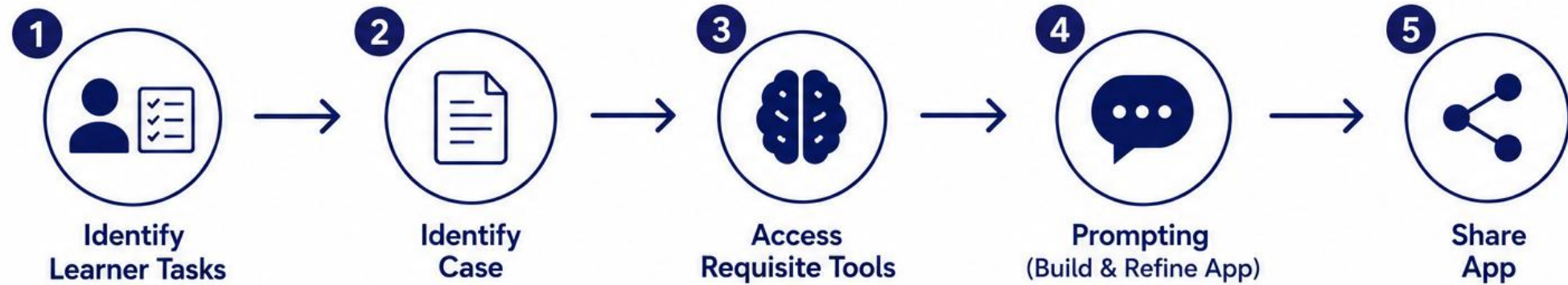
An interactive Shiny application to supplement PR instruction by providing **practice, feedback, reflection, and guided strategies** within time-efficient learning experiences.



Converging evidence from the literature, experts, and learners highlights the need for an accessible, practice-oriented tool to strengthen problem representation skills.

Our App Building Process

Iterate until app meets specified expectations



Lessons Learned from an AI-Assisted Development Process

What we learned and where we're headed

1

CR/PR Instruction Best Practices



- ✓ Frequent feedback
- ✓ Self-explanation
- ✓ Think-alouds
- ✓ Deliberate reflection
- ✓ ...and more

2

Prompting Tool the Most Time— And It Should!



- ✓ Where we shaped task design and AI-supported features
- ✓ Intentional prompting = intentional learning

3

AI Made App Building More Accessible



- ✓ Some programming expertise was needed
- ✓ AI handled most of the heavy lifting, including debugging

4

Shiny + AI as an Educational Development Capability



- ✓ Rapid prototyping of learning tools
- ✓ Iterative educational design
- ✓ Scalable feedback systems
- ✓ Future evaluation and research

Thank You!

References and Resources

